



The hypothesis is clear, specific, and testable, proposing a dual-signal training strategy to achieve zero-shot generalization in a reasonably sized language model. The experimental design is comprehensive and highly rigorous. Experiment 1 directly tests the core claim of zero-shot performance on standard NLU benchmarks (GLUE) against strong and appropriate baselines (GPT-2, GPT-3, InstructGPT). Experiment 2 ensures that the model does not suffer from catastrophic forgetting of its base language generation capabilities by evaluating perplexity on standard datasets (LAMBADA, WikiText2, PTB). Experiments 3 and 4 are well-designed ablation studies that isolate the contributions of the language signal and the task prefix, which are critical components of the proposed method. Experiment 5 is particularly strong as it rigorously tests true zero-shot generalization by explicitly holding out specific tasks during training and evaluating on them, directly addressing the 'generalizing to unseen tasks' claim. Finally, Experiment 6 provides qualitative insights into generation consistency. The metrics used throughout (Matthew's Correlation, Accuracy, F1, Perplexity) are **standard and perfectly suited for the tasks. Overall, the methodology is sound, controlled, and includes all necessary baselines and ablations.**